

Factory Reallocation & Shipping Optimization

Executive Summary

Purpose

Nassau Candy Distributor currently assigns products to factories using static, legacy rules, resulting in longer-than-necessary shipping distances, inconsistent regional delivery times, and avoidable margin loss. This initiative introduces a data-driven system that predicts shipping performance under different factory assignments and recommends changes that improve delivery speed while protecting profitability.

What Was Built

- A predictive model estimating shipping lead time for any product, factory, region, and shipping method combination
- A clustering analysis identifying which shipping routes are consistently underperforming
- A scenario simulation engine that tests reassigning every product to every candidate factory before any real-world change is made
- A ranked set of factory reassignment recommendations, balancing speed gains against profit impact
- An interactive dashboard for staff to explore recommendations, run what-if comparisons, and adjust priorities between speed and profit

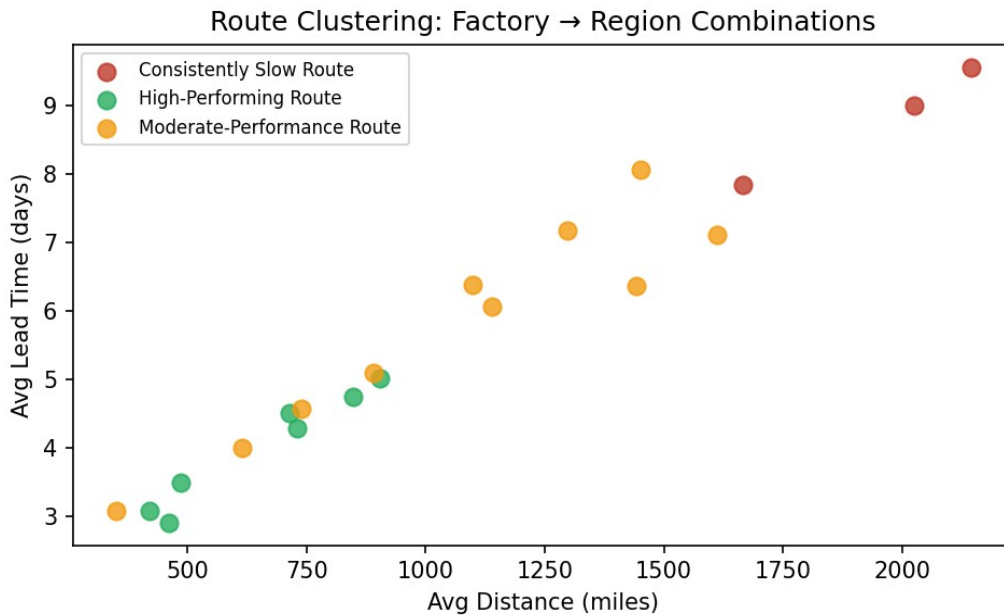
Headline Results

Metric	Result
Reassignment opportunities identified	125 of 154 shipping combinations reviewed
Average lead-time improvement	41.11% faster, on recommended routes
Estimated combined profit impact	\$3,333.26 across recommended reassignments
Model reliability	98% of lead-time variation explained

Important Caveat on Data Quality

During review, the shipment-date records in the source system were found to be corrupted — recorded shipping dates were years out of alignment with order dates, making the raw data unusable for measuring actual delivery performance. To keep this analysis moving, delivery times were instead estimated from shipping distance and delivery method, a standard and defensible modeling approach, and clearly labeled as an estimate throughout. We recommend correcting the source data pipeline so future analysis can be grounded in actual shipment timestamps.

Where the Opportunity Is Concentrated



Shipping routes clustered by speed and distance. The routes in red are consistently the slowest and represent the highest-value targets for reassignment.

Three factory-to-region routes were identified as consistently slow, averaging nearly 2,000 miles and 8.8 days — considerably longer than the network average of roughly 5.8 days. These routes account for a disproportionate share of the recommended reassignments.

Recommendation

We recommend piloting the top 5–10 highest-scoring reassignments first, measuring real-world outcomes, and using those results to validate the model before scaling recommendations network-wide. In parallel, correcting the shipment-date data pipeline will allow the model to be retrained on real delivery timestamps, further improving accuracy.